FERTILITY RATES

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FERTILITY PERIOD

- **Def:** Fertility is the reproductive performance of women. A woman is considered fertile when she has ever given birth to a child **(at least one live birth)**.
- A number of fertility rates (indices) can be calculated, each having particular significance.

1) CRUDE BIRTH RATE (CBR)

- **Crude Birth Rate (CBR)**

= Number of live births in a certain locality and year × 1000

mid year population of the same locality and year

2) GENERAL FERTILITY RATE (GFR)

General Fertility Rate (GFR)

$$= \frac{\text{Number of live births in a certain locality and year}}{\text{1000 Women in the reproductive age 15-49}} \times$$

Advantage:

General fertility rate is more accurate index than the crude birth rate, being related to females in childbearing period (15-49Y) and not to the total population.

CONT. GENERAL FERTILITY RATE (GFR)

- **Drawbacks of GFR**

1 Fertility differences in the age distribution of females aged 15-49 years in population of different countries are not taken into consideration.

2 All females in childbearing period and not the married

- are included.

- **Value of GFR**

- It is usually 4–5 times the birth rate, where females in childbearing period form 20–25 % of the population.

- **GFR:** Like the birth rate, it is generally high in developing countries.

❑ Fecundity rate

Fecundity rate

$$= \frac{\text{Number of live birth in a certain locality \& year}}{\text{Women (married W) in a certain locality \& year}} \times 1000$$

3) AGE-SPECIFIC FERTILITY RATE (ASFR) (AGE-SPECIFIC BIRTH RATE)

Def: It is the number of live births born to mothers of a certain 5-year age group in a given locality and year, per 1000 females of the same age group, locality and year.

Age-specific Fertility Rate (ASFR)

$$= \frac{\text{No of live birth in a certain age group in a certain locality\& year}}{\text{Population of women (in the same age group) in the same locality \& year}} \times 1000$$

ASFR CONT.

Women at childbearing period age groups(7):

- 15-19 Years
- 20-24 Years
- 25-29 Years
- 30-34 Years
- 35-39 Years
- 40-44 Years
- 45-49 Years

CONT.ASFR

- **ASFR is not affected by age distribution thus they can be used for:**
  - Comparisons of fertility behavior at different ages.
  - Comparisons of fertility at different ages over time.
  - Comparisons of fertility across countries / populations.
  - Evaluation of the effectiveness of family planning use on specific age group.

CONT.ASFR

- **Factors affecting ASFR**:

• Marriage pattern e.g. age at marriage.

• Biological factors e.g. menarche, menopause and secondary sterility.

4) TOTAL FERTILITY RATE (TFR)

- Total fertility rate of a population is the average number of children that would be born to a woman over a woman's lifetime if she were to experience the exact current age-specific fertility rates (ASFRs) through her lifetime.

CALCULATION OF TFR

- Total fertility of all females in childbearing period (15- 49 years) can thus be found by adding the age-specific fertility rates of the included 7 groups"5-years each", and multiply by 5.

- It can be expressed per 1000 women or per a woman.

- **TFR**= The age-specific fertility rates (7groups) × 5

5) REPRODUCTION RATE

- Measures the **replacement** of the female population that will sustain the growth of the population.

- **For example**:

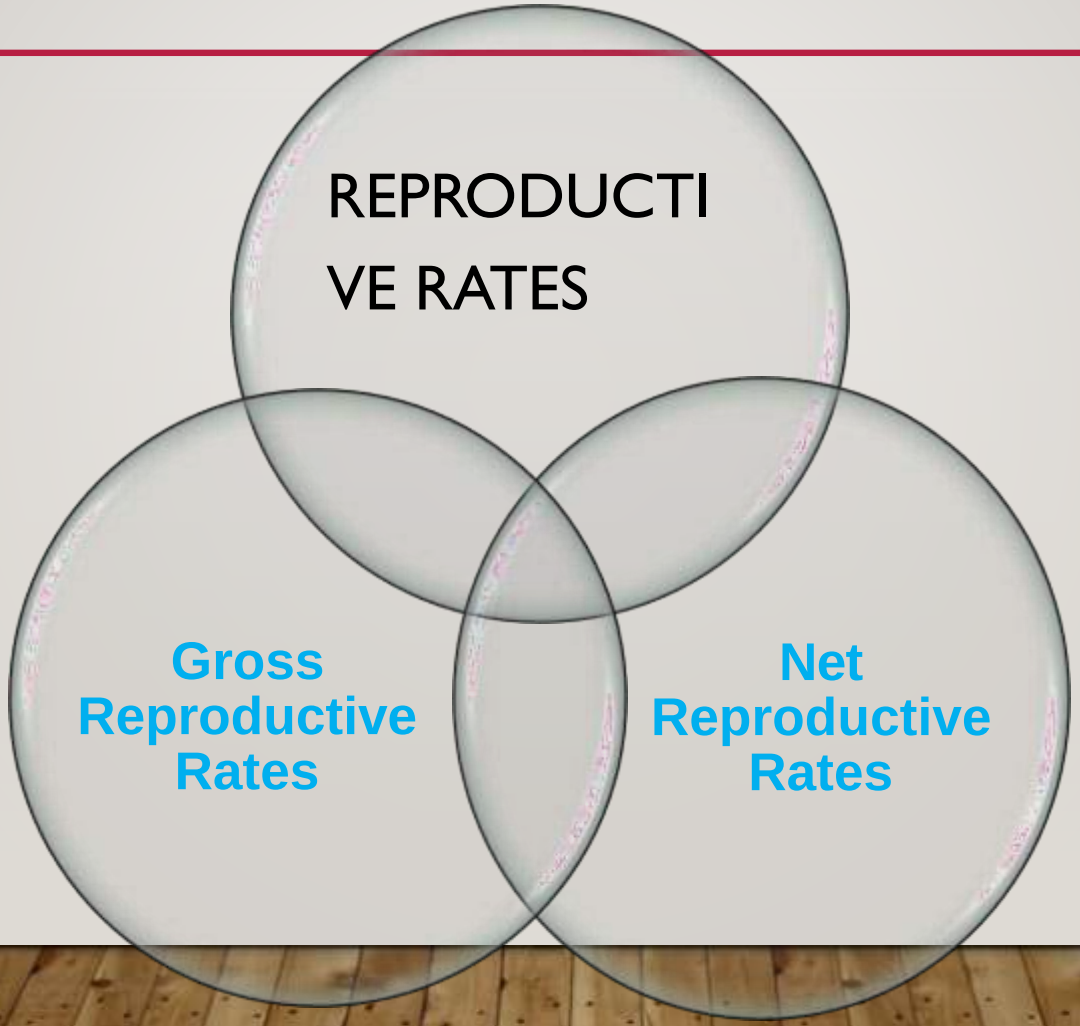
- A Rate of = **1** means exact replacement.
- A Rate **above 1** indicates that the population is more than replacing itself.
- A Rate **below 1** means the population is not replacing itself.

CONT. REPRODUCTION RATE

- Thus, reproduction rate has high correlation with whatever happens to the size of population over time, for example, the higher the number of girl babies, the higher the growth rate of the population and vice versa.

- **Gross and net reproduction rates** used as key indicators for the study of reproduction rate

REPRODUCTIVE RATES



GROSS REPRODUCTION RATE (GRR)

- It is a measure of replacement of mothers by daughters (who are the potential mothers), assuming that all women survive till menopause.

- Gross reproductive rate is calculated by multiplying total fertility rate by the percent of female / total live births.

- **Growth Reproduction Rate GRR= TFR × $\frac{\text{Female live birth}}{\text{Total live births}}$ ×100**

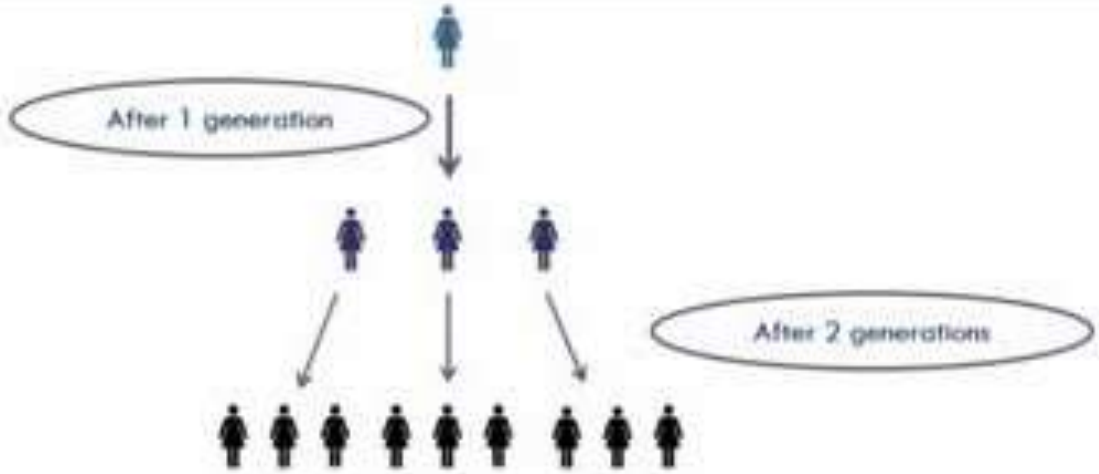
NET REPRODUCTION RATE (NRR)

- This is another measure of replacement of mothers by daughters which takes in consideration that some women will die before menopause.

- **Net reproduction rate (NRR)** = GRR × probability of survival throughout childbearing age.

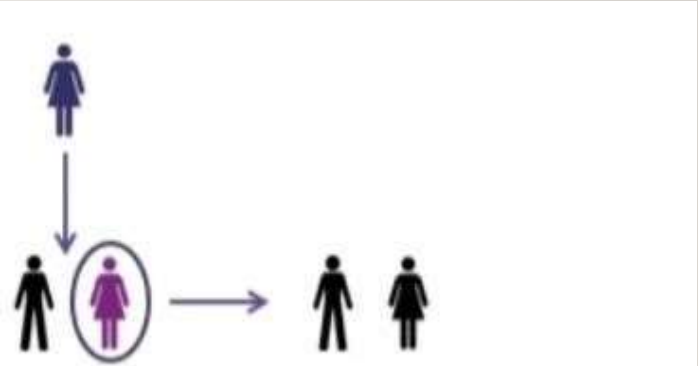
Example:

Reproduction of population when NRR= 3



REPLACEMENT LEVEL FERTILITY

- Level of fertility by which each couple will replaced by another one.
- Replacement level fertility is said to have been reached when:
- NRR =1,0
- GRR >1
- TFR >2 (=2.1).



NRR=1, GRR>1 means that surviving women in the hypothetical cohort have exactly enough daughters (on the average) to replace themselves in the population at this time. TFR > 2 means that the couple has an average of two children.

EXERCISE (1)

- As a health professional you are asked to write a report on fertility in your village; from following table calculate:

a. General fertility rate

b. Total fertility rate

c. Find which age group is the best to be involved in a family planning program?

EXERCISE (1)

Age group	Total population	Female population	Live births
15-19	900	450	45
20 - 24	1000	470	130
25 - 29	900	450	150
30 - 34	600	300	90
35 - 39	680	310	48
40 - 44	470	200	10
45 - 49	450	190	2

ANSWER

A- General fertility rate (GFR)

$$= \frac{\text{Number of live births in a certain locality and year}}{\text{Women in the reproductive age 15-49}} \times 1000$$

$$= \frac{(45+130+150+90+48+10+2)}{(450+470+450+300+310+200+190)} \times 1000$$

$$= \frac{475}{2{,}370} \backslash 1000$$

= 200T1000

ANSWER

B-Total fertility rate (TFR)

Age-specific fertility rates of the included 7 × 5

1) 15-19----------$\frac{45}{450}$ =,1

2) 20 – 24----------$\frac{130}{470}$ = 0,27

3) 25 – 29----------$\frac{150}{450}$ = 0,33

4) 30 – 34 ----------$\frac{90}{300}$ = 0,3

5) 35 – 39---------- $\frac{48}{310}$ = 0,15

6) 40 – 44----------$\frac{10}{200}$ = ,05

7) 45 – 59 --------$\frac{0,2}{190}$ = ,01

TFR=1.21 × 5

=6.05 × 1000

ANSWER

C-The age group is the best to be involved in a family planning program:

which has the highest ASFR?

Age group (25-29)

EXERCISE (2)

1) The numerator for calculating crude birth rate includes

a. All live births

b. All births that survived to their first year of live

c. All live births and still births

d. None of the above

a. All live births

EXERCISE (2)

2) The denominator for general fertility rate relates to

a. All women

b. The total population

c. Women in the reproductive age group

d. Women and men in the reproductive age group

c. Women in the reproductive age group

EXERCISE (2)

3) Summation of age specific fertility rates over all age groups will yield

a. General fertility rate

b. General reproduction rate

c. Total fertility rate

d. Crude birth rate

c. Total fertility rate

EXERCISE (2)

4) The age specific fertility rates (ASFR) for the age groups 15-29 and 30-44 is 300 and 250 respectively. Assuming that there is no fertility before age 15 and after age 45, the total fertility rate for this population will be

a. 6.0

b. 4.0

c. 8.3

d. 2.0

c. 8.3

EXERCISE (2)

5) At replacement levels, the total fertility rate will be

a. Exactly equal to 2

b. Slightly less than 2

c. Slightly more than 2

d. Can be more than or less than 2

c. Slightly more than 2

EXERCISE (2)

- 6) If NRR=1, it implies (means)

a. The crude birth rate is equal to crude death rate

b. The population will stop growing

c. Will be equal to GRR if there is no mortality among women up to 13 years of age

d. Will be equal to GRR if there is no mortality among women up to the end of child bearing years

- **D**

EXERCISE (2)

7) The TFR in a country is 4.8. The sex ratio at birth is 104. The GRR for this country will be

a. 2.35

b. 2.45

c. 3.0

d. 4.2

a. 2.35

EXERCISE (2)

8) The TFR in a country is 3.2. The sex ratio is 100 at birth. There is no mortality among women from birth till the end of child bearing years. The GRR and NRR in this country are

a. 2, 1

b. 1, 2

c. 2, 1

d. 1.6, 1.6

d. 1.6, 1.6

EXERCISE (2)

9) An NRR of 4 means that a woman will be replaced by

a. 16 daughters after 3 generations

b. 64 daughters after 3 generations

c. 4 daughters after 3 generations

d. None of the above.

b. 64 daughters after 3 generations